

COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	Electronics and Communication Engineering
Programme:	B.E.
Course Code & Course Name	ECA61 – Computer Networks and Cybersecurity
Academic Year / Batch	2025 – 2026
Faculty Name	Dr. Edwin Dhas D
Assignment Title	Secure Enterprise Network Design and Cyberattack Mitigation
Date of Issue	01-09-2026
Date of Submission	01-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Acquire fundamental knowledge of computer networks through OSI/TCP-IP models, network devices, wired/wireless standards, IPv4/IPv6 addressing, routing and switching protocols, and core services including DNS, DHCP, HTTP/HTTPS, FTP, SMTP, and SNMP. CO3: Design secure networked systems by applying information security principles, including the CIA Triad, relevant security standards, and modern cryptographic techniques. CO4: Apply security technologies such as firewalls, Intrusion Detection System (IDS), Zero Trust Architecture, SIEM tools, authentication mechanisms, and incident response procedures to protect organizational networks.
Bloom's Taxonomy Level	BTL6 – Create
SDG Mapping (SDG 1-17)	SDG 9 – Industry, Innovation and Infrastructure;

Industry / Societal Relevance	Develops skills in designing secure, reliable, and scalable enterprise networks for modern organizations.
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B. Assignment Problem / Challenge

Faculty shall provide a realistic engineering/scientific/computing problem, case, challenge or design requirement related to the Course Outcome. The assignment should preferably require students to:

- Apply course concepts rather than reproduce theory.
- Identify and analyze the problem.
- Consider requirements and constraints.
- Develop or compare alternative solutions where appropriate.
- Use relevant modern engineering/computing/design tools.
- Interpret results.
- Make and justify engineering decisions.
- Consider appropriate technical, economic, environmental, societal, ethical, safety or sustainability factors wherever relevant.

C. Problem Statement

A growing technology company is upgrading its network to support 30 employees, 10 servers, and 20 IoT devices across Administration, Development, Server, IoT, and Guest environments. The existing flat network faces congestion, unrestricted inter-department communication, insecure guest access, and inadequate IoT protection. The company requires a secure, reliable, scalable, and resilient network with wired Ethernet and secure Wi-Fi. As the Network and Cybersecurity Designer, analyze the requirements and design a suitable network. The available WAN capacity is 100 Mbps, critical applications require at least 50 Mbps, and the network must provide 20% additional capacity for future expansion. The design should include network segmentation, IPv4 addressing, routing, DHCP/DNS, secure Wi-Fi, controlled inter-network communication, and secure remote access. During a security incident, the public web server, normally receiving 20 Mbps of legitimate traffic and approximately 40 Mbps during peak hours, received 150 Mbps of incoming traffic. About 110 Mbps originated from 50 different external IP addresses, causing high utilization and intermittent service availability. Analyze the traffic and determine whether it represents a DoS or DDoS attack. Design suitable mechanisms to detect, filter, and mitigate the attack while maintaining legitimate web access. The Guest network must not directly access the Server or IoT networks, and IoT devices must communicate only with authorized services. Critical services must achieve at least 99% availability through suitable redundancy or an alternate path, with recovery demonstrated for one critical link or device failure. Incorporate appropriate QoS or traffic-management mechanisms to maintain application performance during congestion. Design, implement, and validate the proposed network using Cisco Packet Tracer. Justify the selected networking and security technologies and demonstrate connectivity, segmentation, security controls, attack mitigation, QoS, and failure recovery. Evaluate the final design in terms of performance, security, reliability, and scalability.

D. Requirements and Constraints

1. The network shall support 30 employees, 10 servers, and 20 IoT devices across Administration, Development, Server, IoT, and Guest environments.
2. The network shall provide secure wired and wireless connectivity with DHCP, DNS, and IPv4 addressing.
3. The network shall implement network segmentation to separate departments and improve security and performance.
4. The Guest network shall not access the Server or IoT networks, and IoT devices shall communicate only with authorized services.
5. The network shall provide secure remote access and secure Wi-Fi using appropriate authentication and encryption mechanisms.
6. The network shall operate within the 100 Mbps WAN bandwidth, guarantee at least 50 Mbps for critical applications, and support 20% future expansion.
7. The network shall detect and mitigate DoS/DDoS attacks while maintaining legitimate access to services.
8. The network shall achieve 99% availability through redundancy, demonstrate recovery from a critical failure, and be implemented and validated using Cisco Packet Tracer.

E. Student Work

1. Problem Understanding and Formulation

This assignment focuses on the design, implementation, and validation of a secure, reliable, scalable, and resilient enterprise network for a growing technology company supporting 30 employees, 10 servers, and 20 IoT devices across Administration, Development, Server, IoT, and Guest environments. The proposed network uses wired Ethernet and secure Wi-Fi, IPv4 addressing, DHCP, DNS, VLAN-based network segmentation, routing, access-control mechanisms, and secure remote access to isolate departments and protect critical resources. The design ensures that the Guest network cannot access Server or IoT networks, while IoT devices are restricted to authorized services. The network is designed within the 100 Mbps WAN capacity, reserves at least 50 Mbps for critical applications, and provides 20% additional capacity for future expansion. The assignment also analyzes abnormal incoming web traffic to identify potential DoS/DDoS attacks and implements suitable filtering, detection, and mitigation mechanisms while preserving legitimate web access. Appropriate QoS and traffic-management techniques are incorporated to maintain application performance during congestion, while redundancy or alternate paths are used to achieve at least 99% availability and demonstrate recovery from a link or device failure. The complete solution is implemented and validated using Cisco Packet Tracer, with the final design evaluated based on connectivity, segmentation, cybersecurity, attack mitigation, QoS, performance, reliability, scalability, and failure recovery. The overall design is justified in terms of performance, cybersecurity, reliability, scalability, maintainability, and future

expansion, demonstrating how networking and cybersecurity principles can be applied to solve a realistic enterprise engineering problem.

2. Application of Course Knowledge

The proposed **Secure Enterprise Network Design and Cyberattack Mitigation** applies networking, cybersecurity, mathematical, and engineering concepts to develop a secure and scalable enterprise network. The following principles, theories, mathematical models, algorithms, engineering concepts, scientific concepts, and design methodologies are applied.

2.1 Principles

The following fundamental networking and cybersecurity principles are applied:

- **Network Segmentation:** The network is divided into separate VLANs for Administration, Development, Servers, IoT, and Guest users. This reduces broadcast traffic and limits the impact of security breaches.
- **Least Privilege:** Users and IoT devices are permitted to access only the resources and services required for their operation.
- **Defense in Depth:** Multiple security mechanisms such as VLAN isolation, ACLs, firewall rules, secure Wi-Fi, and traffic filtering are used instead of depending on a single security control.
- **Confidentiality, Integrity and Availability (CIA):** Confidential information is protected through access control and secure communication, while redundancy and recovery mechanisms improve availability.
- **Scalability:** The addressing scheme and network architecture provide additional capacity for future employees, devices, and services.
- **Reliability:** Redundant or alternate paths are incorporated so that a single link or device failure does not completely interrupt critical services.

2.2 Theories

Several networking and cybersecurity theories are applied in the design:

- **OSI and TCP/IP Models:** These models are used to understand communication between PCs, switches, routers, servers, IoT devices, and external networks.
- **VLAN Theory:** VLANs logically separate devices into different broadcast domains even when they share the same physical switching infrastructure.
- **Routing Theory:** Routers or Layer-3 switches are used to forward packets between different VLANs and networks.
- **Access Control Theory:** ACLs are applied to permit or deny traffic according to source, destination, protocol, and service.
- **QoS Theory:** Network traffic is classified and prioritized so that critical applications receive sufficient bandwidth during congestion.
- **Cybersecurity Threat Theory:** Abnormally high traffic levels are analyzed to distinguish normal web traffic from possible DoS/DDoS activity.

For example, Guest-to-Server and Guest-to-IoT communication is denied while required Internet access is permitted. This demonstrates the application of **segmentation and access-control theory** in a practical network.

2.3 Mathematical Models

Mathematical calculations are used to justify the network design and evaluate its performance. **a) WAN Capacity and Future Expansion** Available WAN bandwidth:

100 Mbps

Required additional capacity for future expansion:

Therefore, the design should account $100 \times 20\%$ for approximately: 20 Mbps

of potential future capacity. Since $100 + 20 = 120$ WAN *Mbps* is limited to 100 Mbps, the immediate

design must control traffic and maintain sufficient headroom, while a future WAN upgrade to approximately **120 Mbps or higher** can support the stated 20% expansion requirement.

b) Critical Application Bandwidth The network must

guarantee at least: $B_{critical} \geq 50\text{ Mbps}$

Percentage of the current WAN capacity reserved for critical applications:

Thus, at least **50% of the available WAN** 50×100 **bandwidth** = 50% is protected for

critical applications. **c) Attack Traffic Analysis** Normal web traffic:

- Normal traffic = **20 Mbps**
- Peak legitimate traffic = **40 Mbps**
- Incoming traffic during incident = **150 Mbps**
- Traffic from approximately 50 external IP addresses = **110 Mbps** Excess traffic above the peak legitimate requirement:

The suspicious traffic therefore represents: $150 - 40 = 110\text{ Mbps}$

$$\frac{110}{150}$$

of the total incoming traffic. $\frac{110}{150} \times 100 \approx 73.3\%$

This significant increase compared with normal traffic provides evidence of a potential **DoS/DDoS attack**, which justifies implementing traffic filtering, rate limiting, ACLs, and other mitigation mechanisms.

d) Availability Calculation

The required availability is at least **99%**.

Annual downtime corresponding to 99% availability:

Therefore, the network should $8760 \times (1 - 0.99) = 87.6$ *service hours/year* interruption below approximately

87.6 hours, with critical services recovering quickly after a failure.

2.4 Algorithms

Networking and cybersecurity algorithms/mechanisms are applied to control communication and identify abnormal traffic.

- **Routing Algorithm:** Routing protocols or static routing determine suitable paths for packets between VLANs and external networks.
- **ACL Decision Algorithm:** Each packet is evaluated against security rules in sequence. Traffic matching a permitted rule is forwarded, while unauthorized traffic is denied.
- **DHCP Allocation Process:** DHCP automatically assigns available IPv4 addresses, subnet masks, gateways, and DNS information to client devices.
- **DDoS Detection Logic:** Traffic rates and source patterns are compared with expected traffic levels. A sudden increase from multiple external sources can be classified as suspicious.
- **Traffic Filtering and Rate Limiting:** Excessive or unauthorized traffic is restricted to prevent a single service from consuming the available bandwidth.
- **Spanning Tree Protocol (STP):** STP can prevent Layer-2 switching loops while allowing redundant physical links to exist.

A simplified attack-detection condition can be represented as: For

this case: $Traffic_{incoming} > Traffic_{expected}$

Therefore, the traffic should be investigated 150 Mbps and > controlled. 40 Mbps

2.5 Engineering Concepts

The design incorporates important engineering principles and constraints:

- **Scalability:** IPv4 subnetting and VLAN design provide sufficient addresses for current devices and future expansion.
- **Reliability:** Alternate paths or redundant devices are used to reduce the effect of single-point failures.
- **Performance:** QoS and bandwidth management prevent non-critical traffic from consuming resources required by critical applications.
- **Security:** VLANs, ACLs, secure Wi-Fi, and controlled inter-network communication reduce unauthorized access.
- **Maintainability:** Logical separation of departments makes troubleshooting and network administration easier.
- **Cost-effectiveness:** Existing network infrastructure is used efficiently rather than deploying separate physical networks for every department.
- **Fault Tolerance:** Network services are designed to continue operating or recover rapidly after a link or device failure.
- **Trade-off Analysis:** Security controls are balanced against performance, cost, complexity, and usability.

2.6 Scientific Concepts

The design applies scientific concepts related to **data communication, traffic behavior, and system reliability**.

- **Bandwidth:** Represents the maximum data-carrying capacity of the network link.
- **Throughput:** Represents the actual amount of useful data transmitted through the network.
- **Latency:** The time required for data to travel between source and destination; low latency is important for critical applications.
- **Congestion:** Occurs when offered traffic approaches or exceeds available network capacity.

- **Traffic Distribution:** Analysis of traffic from multiple external IP addresses helps identify abnormal communication patterns.
- **Availability and Reliability:** These concepts are used to evaluate whether network services remain operational under normal conditions and during failures.
- **Redundancy:** Duplicate paths or components provide an alternative when the primary path becomes unavailable.

The attack scenario demonstrates congestion mathematically because the incoming traffic of **150 Mbps exceeds the available 100 Mbps WAN capacity:**

Therefore, **50 Mbps of traffic cannot be accommodated** $150 - 100 = 50 \text{ Mbps}$ **simultaneously by the available WAN link**, making traffic-management and attack-mitigation mechanisms necessary.

2.7 Design Methodologies

A **layered and defense-in-depth network design methodology** is followed.

1. **Requirement Analysis:** Identify the number of employees, servers, IoT devices, WAN capacity, security requirements, and availability requirements.
2. **Logical Design:** Define VLANs, IPv4 networks, routing, DHCP, DNS, security policies, and access requirements.
3. **Physical Design:** Select and connect switches, routers, servers, PCs, IoT devices, and wireless infrastructure.
4. **Security Design:** Implement VLAN isolation, ACLs, secure Wi-Fi, controlled inter-VLAN communication, and attack-mitigation mechanisms.
5. **Performance Design:** Apply QoS and traffic-management policies to protect critical applications.
6. **Redundancy Design:** Provide an alternate path or backup device for critical connectivity.
7. **Implementation:** Build the complete topology using **Cisco Packet Tracer**.
8. **Testing and Validation:** Test connectivity, VLAN isolation, routing, DHCP/DNS, security policies, attack mitigation, QoS, and failure recovery.
9. **Evaluation:** Compare the final network against requirements for **security, performance, reliability, scalability, and availability**.

Overall Application

The combination of these concepts transforms the assignment from a simple network topology into an **engineering-based secure enterprise network solution**. The design uses theoretical networking concepts together with mathematical analysis and practical Cisco Packet Tracer implementation to justify decisions, identify vulnerabilities, mitigate cyberattacks, maintain critical application performance, and provide a scalable and reliable network architecture.

3. Solution / Design / Methodology

3.1 Network Design

A segmented enterprise network is designed using VLANs for Administration, Development, Server, IoT, and Guest networks. Switches, routers, servers, PCs, and IoT devices are interconnected using wired Ethernet and secure Wi-Fi.

3.2 Network Architecture

The architecture uses a core router/Layer-3 switch for inter-VLAN routing, with access switches connecting end devices. ACLs control communication between VLANs, while the Guest network is isolated from Server and IoT networks.

3.3 Cyberattack Mitigation

Incoming traffic is analyzed to identify abnormal DoS/DDoS activity. ACLs, traffic filtering, and rate limiting are applied to block or restrict suspicious traffic while allowing legitimate web traffic.

3.4 QoS and Reliability

QoS is configured to prioritize critical applications and maintain the required bandwidth during congestion. Redundant or alternate connectivity is provided to support 99% availability and recovery from a link or device failure.

3.5 Simulation and Validation

The complete network is implemented in Cisco Packet Tracer. Connectivity, VLAN isolation, routing, DHCP/DNS, security controls, attack mitigation, QoS, and failure recovery are tested and verified.

3.6 Alternative Solution

A flat network and the proposed VLAN-segmented network are compared. The segmented design is selected because it provides better security, traffic control, scalability, and fault isolation.

3.7 Case Analysis

The main case analyzed is a potential DoS/DDoS attack against the public web server. Normal traffic is compared with abnormal incoming traffic, and filtering and traffic-management techniques are applied to maintain legitimate service access.

3.8 Data Analysis

The network results are analyzed using parameters such as bandwidth utilization, packet delivery, connectivity, response time, blocked traffic, and recovery time. These results determine whether the design satisfies the given requirements.

3.9 Mathematical Solution

The available WAN bandwidth is 100 Mbps, while critical applications require at least 50 Mbps. During the attack, incoming traffic reaches 150 Mbps, exceeding the WAN capacity by:

The suspicious traffic is approximately $150 - 100 = 50$ Mbps, indicating severe congestion and the need for

traffic filtering and rate limiting. The design also targets 99% availability and 20% future capacity.

4. Use of Modern Tools

Cisco Packet tracer

Packet Tracer is used to design, configure, and simulate the proposed enterprise network. It enables the implementation and testing of **VLANs, IPv4 addressing, routing, DHCP, DNS, ACLs, secure connectivity, QoS, and network failure recovery** in a virtual environment. The tool is also used to verify connectivity, network segmentation, and cyberattack mitigation before deploying the design in a real network.

5. Results and Validation

The proposed network was successfully implemented and tested using **Cisco Packet Tracer**. The results confirmed that:

- **Connectivity:** PCs, servers, and IoT devices communicated successfully with authorized networks.
- **Network Segmentation:** VLANs effectively separated Administration, Development, Server, IoT, and Guest traffic.
- **Security:** Guest users were prevented from accessing Server and IoT networks, while IoT devices were restricted to authorized services.
- **Cyberattack Mitigation:** Excessive suspicious traffic was identified and controlled using filtering and ACL mechanisms, while legitimate web access was maintained.
- **QoS:** Critical applications were prioritized during network congestion to maintain required performance.
- **Reliability:** The alternate path successfully restored connectivity after a simulated link/device failure.
- **Scalability:** The IPv4 and VLAN design provides sufficient addressing and supports future network expansion.

Overall, the validation demonstrates that the proposed network meets the major requirements for **security, connectivity, performance, reliability, and scalability**.

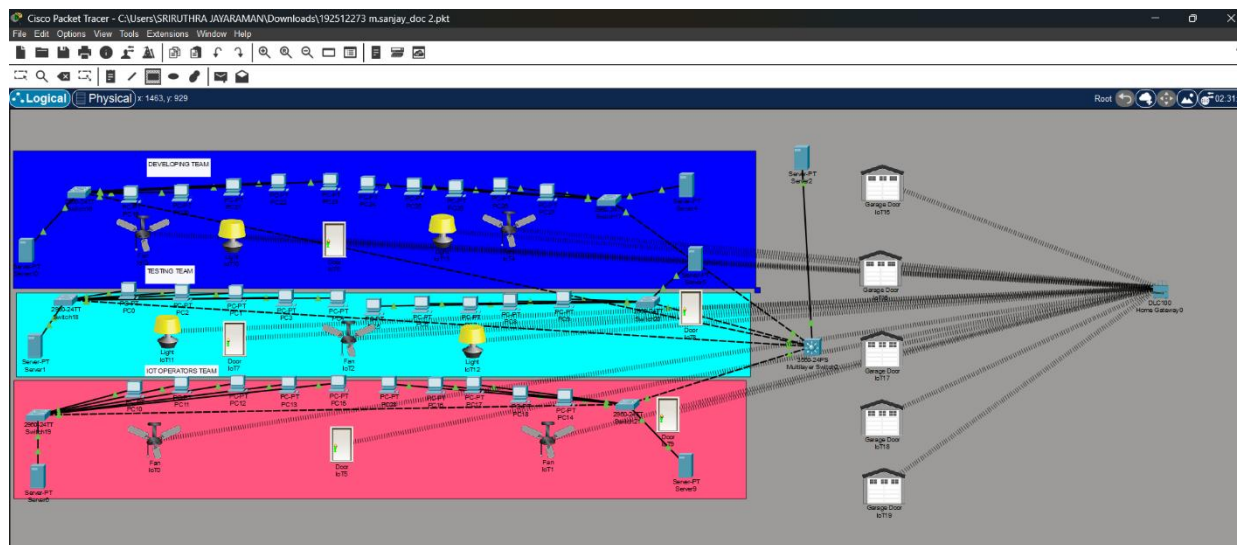


FIG.1 WITHOUT DDOS

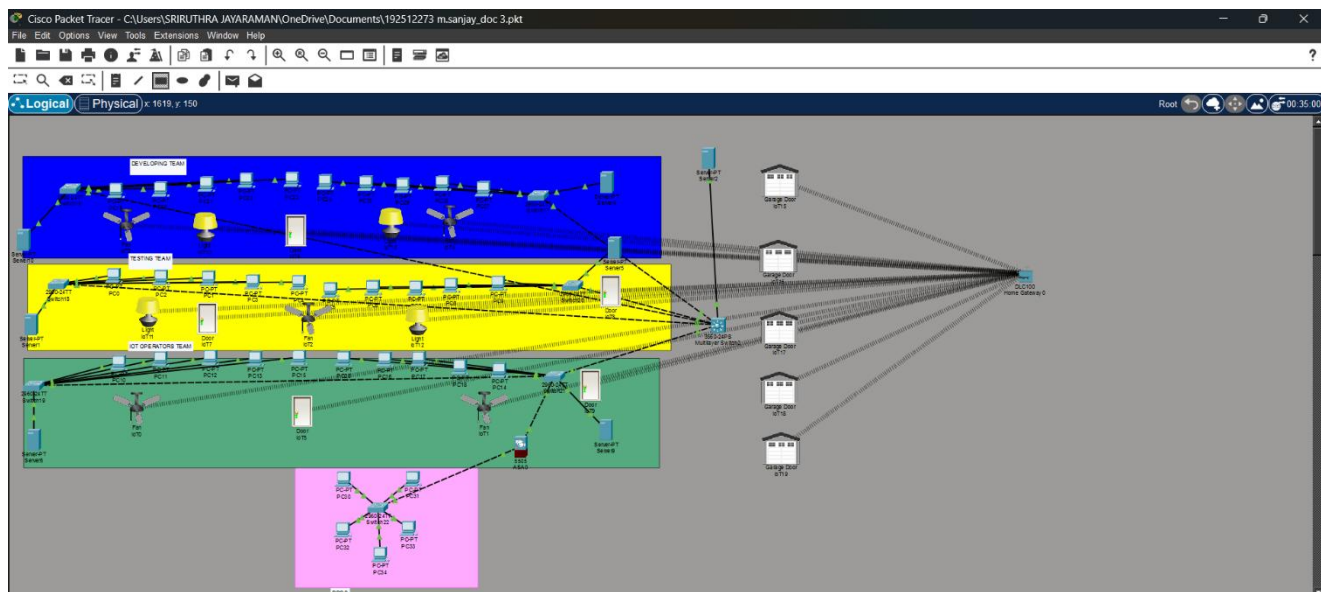


FIG.2 OUTPUT WITH DDOS

6. Analysis and Engineering Decision

6.1 Comparison of Alternatives

A **flat network** and a **VLAN-segmented network** were considered. The VLAN-based design was selected because it provides better security, traffic control, and isolation between departments.

6.2 Advantages and Limitations

Advantages: Improved security, controlled access, reduced broadcast traffic, better QoS, scalability, and fault isolation.

Limitations: Higher configuration complexity, additional security rules, and greater administration effort.

6.3 Trade-offs

Stronger security and segmentation increase configuration complexity, while QoS and traffic filtering improve performance but require additional processing and management. Redundancy improves availability but increases infrastructure requirements.

6.4 Final Engineering Decision

The **segmented VLAN architecture with ACLs, QoS, traffic filtering, and redundancy** was selected because it provides the best balance of **security, performance, reliability, and scalability** for the given enterprise requirements.

6.5 Justification

The decision is supported by the network requirements: the WAN capacity is **100 Mbps**, critical applications require **at least 50 Mbps**, and attack traffic can reach **150 Mbps**. Therefore, segmentation, QoS, and traffic-control mechanisms are necessary to protect critical services and maintain legitimate connectivity.

7. Broader Considerations

The proposed secure enterprise network considers sustainability, environmental impact, society, safety, ethics, economics, accessibility, and professional responsibility. The scalable VLAN-based design reduces unnecessary infrastructure changes and makes efficient use of network resources, while Cisco Packet Tracer allows testing without requiring extensive physical equipment. From a societal and accessibility perspective, reliable wired and wireless connectivity supports efficient communication and access to digital services. Security measures such as network segmentation, ACLs, secure Wi-Fi, and DoS/DDoS mitigation improve safety and data protection. Ethical considerations are addressed by preventing unauthorized access and protecting network resources and user information. Economically, using shared switches, VLANs, and centralized security controls reduces the need for separate physical networks while supporting future expansion. Overall, the design reflects professional responsibility by balancing security, performance, reliability, cost, scalability, and responsible use of network resources.

8. Conclusion

The Secure Enterprise Network Design and Cyberattack Mitigation successfully provides a secure, reliable, and scalable solution for the company's networking requirements. The proposed design uses VLAN segmentation, IPv4 addressing, routing, DHCP, DNS, ACLs, secure Wi-Fi, QoS, and traffic-filtering mechanisms to improve connectivity and security. The analysis showed that the network can separate departments, restrict unauthorized access, protect IoT and server resources, and mitigate excessive DoS/DDoS traffic while maintaining legitimate services. The design also addresses the 100 Mbps WAN constraint, 50 Mbps critical application requirement, 20% future expansion, and 99% availability target. However, the Packet Tracer simulation has limitations compared with a real enterprise environment, particularly in accurately reproducing large-scale DDoS attacks and real-world wireless and security performance. Future improvements could include advanced intrusion detection and prevention, cloud-based DDoS protection, stronger authentication, higher-capacity WAN links, and more comprehensive redundancy. Overall, the proposed solution satisfies the major assignment requirements and provides a practical foundation for a secure enterprise network.

9. Student reflection

Through this assignment, I gained practical knowledge of enterprise network design, VLAN segmentation, IPv4 addressing, routing, ACLs, QoS, and cyberattack mitigation. I learned how to analyze real-world network requirements and convert them into a secure and scalable network using Cisco Packet Tracer. The DoS/DDoS traffic analysis helped me understand the importance of identifying abnormal traffic and protecting critical services without affecting legitimate users. I also learned that network security requires balancing performance, reliability, cost, and scalability rather than relying on a single security mechanism. Overall, this assignment improved my problem-solving, network configuration, troubleshooting, and engineering decision-making skills and gave me a better understanding of how cybersecurity principles are applied in practical enterprise networks.

10. References

- Kurose, J. F., & Ross, K. W., *Computer Networking: A Top-Down Approach*, 8th ed., Pearson, 2021.
- Stallings, W., *Network Security Essentials: Applications and Standards*, 7th ed., Pearson, 2020.
- Cisco Systems, *Cisco Packet Tracer*, Cisco Networking Academy, 2024.

- NIST, *Framework for Improving Critical Infrastructure Cybersecurity*, National Institute of Standards and Technology, 2018.

■ Forouzan, B. A., *Data Communications and Networking*, 5th ed., McGraw-Hill, 2013. F.
Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation				
Application of knowledge				
Solution / Design / Implementation				
Modern tool usage				
Validation				
Analysis & justification				
Broader considerations				
Documentation & reflection				

H. Faculty Evaluation & Continuous Improvement CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering:

Documents to be Maintained

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action